

Fine-tuning a Tsuboyama-pretrained $\Delta\Delta G$ model on FireProt

Experiment 08 · ddG_with_Boltz · concat features + antisymmetry · MLP · cross-dataset homology holdout.

A $\Delta\Delta G$ regressor pretrained on Tsuboyama and then **sequentially fine-tuned on FireProt** improves FireProt accuracy — Spearman +0.03–0.05 across every homology threshold — while losing essentially nothing on Tsuboyama (≤ 0.012 Pearson). Fine-tuning transfers extra signal from the second dataset without catastrophic forgetting; the effect is modest, bounded by the same feature ceiling seen in the transfer study.

1. Motivation

A Tsuboyama-trained model transfers to the independently curated FireProt dataset but is bounded by how densely the training data covers each $\Delta\Delta G$ value. A natural way to add coverage is to fine-tune on FireProt itself. The question is whether this helps FireProt, and whether it degrades the original Tsuboyama performance (catastrophic forgetting) — measured on both datasets under a split that controls for homology across them.

2. Methods

Splits. All wild-type sequences from both datasets are pooled and clustered by sequence identity (MMseqs2, 80% coverage) at 30/50/90%. Whole clusters are assigned to train or test, so no train/test pair — within or across datasets — exceeds the threshold; clusters spanning both datasets go to train. This yields disjoint *Tsuboyama-train* / *Tsuboyama-test* / *FireProt-finetune* / *FireProt-test* sets.

Model. 5-seed MLP ensemble on concat pair-track features with antisymmetry augmentation (the defaults established in experiment 07). **A (Tsuboyama-only):** trained on Tsuboyama-train. **D (fine-tuned):** the same model, warm-start continued on FireProt-finetune at a lower learning rate; the input normalisation is fixed from pretraining. Both are evaluated on Tsuboyama-test and FireProt-test.

3. Results

Identity	Pearson r	Spearman ρ	RMSE
30%	0.466 → 0.522	0.693 → 0.739	2.13 → 1.97
50%	0.507 → 0.528	0.672 → 0.716	1.93 → 1.80
90%	0.355 → 0.343	0.658 → 0.686	1.78 → 1.84

Table 1. FireProt-test — A (Tsuboyama-only) → D (fine-tuned). Bold where D improves.

Identity	Pearson r (A → D)
30%	0.794 → 0.790
50%	0.787 → 0.784
90%	0.790 → 0.778

Table 2. Tsuboyama-test Pearson r — A → D (forgetting check).

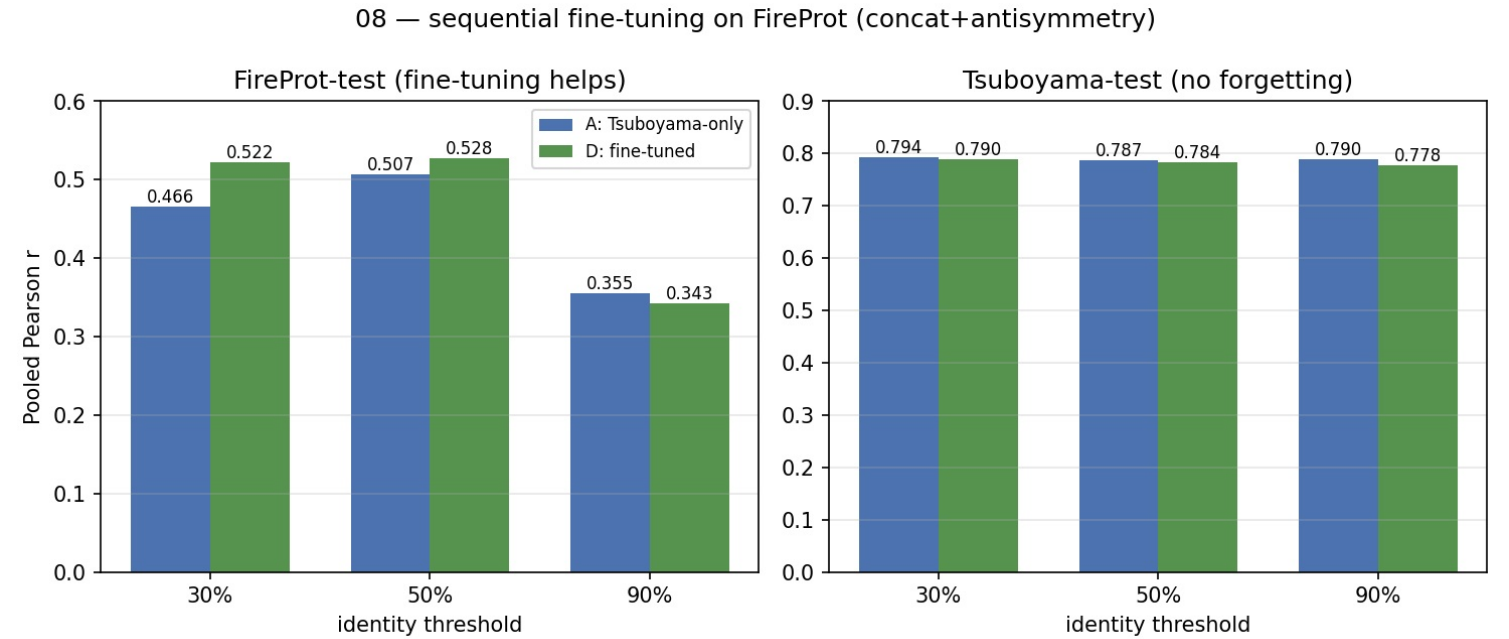


Figure 1. A vs D pooled Pearson r. Left: FireProt-test (fine-tuning helps at 30/50%). Right: Tsuboyama-test (unchanged — no forgetting).

4. Discussion

Fine-tuning gives a **modest, consistent FireProt gain** — Spearman rises at all three thresholds and Pearson/RMSE improve at 30% and 50% — with **negligible Tsuboyama forgetting** (≤ 0.012 Pearson). At the strictest 90% split the FireProt-test set is smallest and noisiest;

there Pearson dips slightly while Spearman still rises, a calibration wobble rather than a loss of ranking signal. The overall size of the effect is limited: the predictor's accuracy is set largely by the pair-track features, so adding FireProt labels refines rather than transforms it. Sequential fine-tuning is a safe, mild improvement when a second labelled dataset is available.